

**BIU LOST AND FOUND MANAGEMENT SYSTEM FOR BELTEI
INTERNATIONAL UNIVERSITY**

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NHANH KIMSON

2026

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INTERNATIONAL UNIVERSITY**

THE BACHELOR'S THESIS IN PARTIAL FULFILLMENT
OF THE REQUIREMENT FOR THE DEGREE OF BACHELOR
OF INFORMATION TECHNOLOGY IN SOFTWARE ENGINEERING

**BIU LOST AND FOUND MANAGEMENT SYSTEM FOR BELTEI
INTERNATIONAL UNIVERSITY**

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2026

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COMMITTEE APPROVAL

This Bachelor's Thesis entitled "BIU Lost and Found Management System for Beltei International University" was prepared and submitted by NHANH KIMSON of the Beltei International University in partial fulfilment of the requirement of a Bachelor of Software Engineering (BIT-SE).

Approved by the University Evaluation Committee with a grade of PASSED

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DECLARATION

I do hereby declare that, except otherwise stated the Bachelor's Thesis "BIU Lost and Found Management System for Beltei International University" based on my original work and the same has not been submitted either in part or in full for the award of any other degree of this or any other University.

My indebtedness to other writer(s) has/have been acknowledged at relevant places.

NHANH KIMSON

Date Signed

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ABSTRACT

Lost and found management is a critical campus service that directly affects student satisfaction, operational efficiency, and the safe return of personal belongings. For Beltei International University, this study investigates the development of a BIU Lost and Found Management System (BLFMS) with the goal of improving the speed, transparency, and reliability of reporting lost items, publishing found items, matching potential owners, and processing ownership claims. Many universities still rely on bulletin boards, informal social media posts, or fragmented manual records, which frequently result in delayed reunions, duplicate reports, and poor visibility of open cases. This study examines the challenges faced by students and campus staff in lost-and-found coordination and proposes a digital solution that integrates structured item reporting, image uploads, searchable listings, automated match suggestions, claim verification, in-app notifications, and role-based administration.

Through a mixed-method approach combining surveys, interviews, and direct observation of existing campus practices, the study identifies key weaknesses in manual coordination and evaluates user readiness for a centralized web platform. The proposed system is designed with a user-friendly interface, secure authentication, validated forms, and a scalable PostgreSQL database to ensure accuracy, accessibility, and maintainability. Built with Next.js, Prisma, NextAuth, Cloudinary, and REST APIs documented in OpenAPI, the system supports both browser users and external clients.

The findings indicate that implementing BLFMS can significantly reduce the time required to locate matching listings, improve claim traceability, and increase confidence in campus lost-and-found processes. The system also supports better decision-making for administrators through dashboards, claim review tools, and activity analytics. Overall, the research demonstrates that digital transformation in campus lost-and-found services is both technically feasible and strongly supported by end users.

Keyword: Lost and Found, Campus Information System, Digital Transformation, Web Application, Claim Management, Beltei International University, Software Engineering.

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LIST OF ABBREVIATIONS

BLFMS	: BIU Lost and Found Management System
BIU	: Beltei International University
API	: Application Programming Interface
REST	: Representational State Transfer
ORM	: Object-Relational Mapping
UI	: User Interface
UX	: User Experience
TAM	: Technology Acceptance Model
RBV	: Resource-Based View
JWT	: JSON Web Token
SSE	: Server-Sent Events
HTTP	: Hyper Text Transfer Protocol
HTTPS	: Hyper Text Transfer Protocol Secure
JSON	: JavaScript Object Notation
SQL	: Structured Query Language
CRUD	: Create, Read, Update, Delete
KPI	: Key Performance Indicator
GDPR	: General Data Protection Regulation

CHAPTER 1: INTRODUCTION

1.1 Introduction to Research

Effective campus service management is essential for universities seeking to improve student experience, operational transparency, and institutional reputation. Lost and found services are a routine but high-impact part of campus life because students and staff frequently misplace valuable items such as laptops, student IDs, keys, bags, and personal documents. When these belongings are not recovered quickly, the consequences include academic disruption, financial loss, stress, and reduced trust in campus support systems.

At Beltei International University, lost-and-found coordination has traditionally depended on informal channels such as front-desk notices, printed lists, word of mouth, and social media posts. Although these methods can work in limited cases, they are difficult to search, hard to audit, and vulnerable to incomplete or outdated information. As campus populations grow and digital expectations increase, universities need structured systems that centralize reports, preserve evidence such as photos and location details, and support verification before items are returned.

The BIU Lost and Found Management System (BLFMS) proposed in this study addresses these needs by providing a web-based platform where users can report lost or found items, browse and filter listings, receive match suggestions, submit ownership claims with proof, and track notifications from a personal dashboard. Administrators and staff can review claims, manage listings, monitor activity, and maintain accountability across the full item lifecycle.

This research is grounded in software engineering practice and supported by established theories such as the Technology Acceptance Model (TAM), Resource-Based View (RBV), and Institutional Theory. These frameworks help explain why users adopt digital campus systems, how technology improves service efficiency, and why institutions invest in standardized platforms to meet operational and compliance expectations.

1.2 Research Problem

Although lost-and-found services are important, Beltei International University currently faces challenges related to fragmented reporting, poor searchability, delayed communication, and limited claim traceability. When a student loses an item, they may

not know where to report it, whether anyone else has already found it, or how to prove ownership later. Similarly, finders may post information in one channel while owners search in another, reducing the chance of successful reunions.

Manual and decentralized practices also make it difficult for staff to monitor open cases, approve claims consistently, or measure service performance. These weaknesses increase administrative workload, create uncertainty for users, and reduce the overall effectiveness of campus support. Therefore, this research addresses the lack of a comprehensive, integrated system that can manage the full lost-and-found workflow at Beltei International University.

1.3 Research Aim/Objective

The main goal of this research is to design, implement, and evaluate BLFMS to improve the speed, accuracy, and transparency of campus lost-and-found management. The specific objectives include:

- Identifying the key components and functionalities required for a successful BLFMS.
- Examining the impact of BLFMS on item recovery efficiency and claim traceability.
- Assessing the system's role in improving communication between students, finders, and staff.
- Analyzing the advantages and challenges of BLFMS compared to manual lost-and-found methods.
- Conducting a literature review on campus information systems and digital service management.
- Identifying key challenges faced by users in reporting, searching, and claiming items.
- Evaluating user readiness for adopting a centralized lost-and-found web platform.
- Providing recommendations for sustainable deployment and future enhancement.

1.3.1 Research Question

The primary research questions driving this study are:

- What are the common challenges students and staff face in managing lost and found items on campus?
- How can a digital system improve the speed and accuracy of matching lost and found reports?
- What measurable benefits does BLFMS provide for users, administrators, and campus operations?

1.3.2 Signification of the study

The significance of this study lies in its contribution to the digital transformation of campus support services. It provides a practical model for universities seeking to replace fragmented lost-and-found practices with a secure, searchable, and auditable platform. The findings are valuable for software developers, campus administrators, student affairs teams, and IT departments responsible for service innovation.

1.4 Scope and limitation

The study focuses on the analysis, design, implementation, and evaluation of BLFMS at Beltei International University. The scope covers lost item reporting, found item reporting, image uploads, search and filtering, match suggestions, claim submission and review, notifications, user dashboards, and administrative monitoring.

1.4.1 Scope and Limitations

This study examines lost-and-found management only within the Beltei International University campus context. Data were collected from students, staff, and administrators using questionnaires, interviews, and observation. The system is implemented as a web application using Next.js, Prisma, PostgreSQL, NextAuth, Cloudinary, and REST APIs. The research emphasizes improvements in reporting quality, searchability, claim verification, and service transparency.

1.4.2 Limitations of the Study

Despite careful planning, this study has several limitations. First, the research was conducted only at Beltei International University, which may limit generalization to other universities. Second, the sample size is relatively small and focused on campus users directly involved in or affected by lost-and-found processes. Third, the study relies partly on self-reported survey data, which may be influenced by perception bias. Fourth, time constraints limited long-term evaluation of recovery rates after deployment. Finally, confidentiality and privacy requirements restricted access to some operational records.

1.5 Layout of the study/Research

CHAPTER 1 INTRODUCTION introduces the background, research problem, objectives, research questions, significance, scope, and limitations.

CHAPTER 2 LITERATURE REVIEW reviews definitions, theories, prior studies, and the conceptual framework for BLFMS.

CHAPTER 3 RESEARCH METHODOLOGY describes the research design, location, population, sampling, UI/UX scope, and data collection instruments.

CHAPTER 4 DATA ANALYSIS analyzes strengths and weaknesses of current practices and proposes BLFMS as the solution.

CHAPTER 5 RESEARCH FINDINGS AND DISCUSSIONS presents findings and discusses their implications for campus service performance.

CHAPTER 6 CONCLUSION AND RECOMMENDATIONS summarizes the study and provides practical recommendations for deployment and improvement.

CHAPTER 2: LITERATURE REVIEW

2.1 Definitions and Theories Related to the Topic

Lost and found management refers to the organized process of recording, publishing, searching, matching, verifying, and returning misplaced belongings within an institution. In a university context, this service connects reporters, finders, owners, and administrators through structured information about item type, category, location, date, description, and proof.

A BIU Lost and Found Management System (BLFMS) is an integrated digital platform that automates these tasks through web forms, searchable listings, image storage, claim workflows, notifications, and administrative review tools. Such systems improve consistency, reduce duplicate effort, and create an auditable record of each case.

Relevant theories include:

- Resource-Based View (RBV): campus service systems are strategic resources that improve institutional efficiency and user satisfaction.
- Technology Acceptance Model (TAM): adoption depends on perceived usefulness and ease of use.
- Institutional Theory: universities adopt standardized systems to meet service expectations, accountability, and operational norms.

2.2 Analysis of Scholars' Concepts/Words/Sayings

- Traditional lost-and-found methods are prone to information loss, poor searchability, and delayed communication.
- Integrated campus platforms improve transparency, reduce administrative burden, and support evidence-based claim decisions.
- Digital service systems can standardize workflows while still allowing role-based flexibility for students and staff.
- User-centered design and validation are critical for accurate reporting and trustworthy claim processing.

2.3 Experiences or Issues Raised

- Implementation challenges include change resistance, training needs, and integration with existing campus processes.
- Data security and privacy are essential because item reports may include contact details and proof images.

- User acceptance and training determine whether a new system is used consistently or bypassed in favor of informal channels.
- Successful deployment improves recovery speed, claim traceability, and confidence in campus support services.

2.4 Research Model or Framework

The research model illustrates how current lost-and-found practices influence organizational outcomes through the implementation of BLFMS. The framework contains independent variables, an intervening variable, and dependent variables.

Independent variables include reporting accuracy, search efficiency, communication speed, workflow automation, and claim verification quality. These feed into BLFMS as the intervening system, which provides structured reporting, image evidence, match suggestions, notifications, and admin review. Dependent variables include faster item recovery, improved claim traceability, reduced duplicate reports, higher user satisfaction, and better administrative control.

Figure 1. Conceptual framework for BLFMS

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Research Methodology

This study used a mixed-method approach at Beltei International University to understand real-world lost-and-found challenges and evaluate the need for BLFMS. Qualitative methods included interviews and observation. Quantitative methods included structured questionnaires distributed to students and staff.

3.1.1 Research Design

A descriptive and analytical research design was adopted. The descriptive component documented current practices and user experiences. The analytical component evaluated weaknesses and proposed an automated web-based solution.

3.1.2 Research Location

The research was conducted at Beltei International University in Phnom Penh, Cambodia. Phnom Penh is the capital and a major center for higher education and technology adoption, making it a suitable environment for studying campus digital services.

Figure 2. Map of Cambodia

3.1.3 Research Population

Department	Total Population	Proportional Allocation (Sample Size)
Student Affairs / Front Desk	5	5
Students	20	20
Total	30	30

Table 1 List of participation

3.1.4 Research Data Scope of UI

The UI scope of BLFMS focuses on a clear, mobile-friendly interface for reporting, browsing, claiming, and administering lost-and-found cases.

Figure 3 Dashboard BLFMS

Backend and platform technologies:

- Next.js App Router: full-stack React framework for server components, routing, and API routes.
- Prisma ORM: type-safe database access for Item, Claim, User, and Notification models.

- PostgreSQL (Neon): scalable relational storage for listings, claims, sessions, and audit data.
- NextAuth v5: secure authentication with credentials and optional social login.
- Cloudinary: image upload, storage, and delivery for item and proof photos.
- Zod + React Hook Form: validated multi-step forms for lost and found reporting.
- OpenAPI + Swagger UI: documented REST API for web and external clients.

Figure 4 Login Page

Figure 5 Report Lost Item Page

Figure 6 Report Found Item Page

Figure 7 Item Detail and Claim Page

Figure 8 Admin Claims Review Page

Frontend technologies:

- React 19 + TypeScript strict mode
- Tailwind CSS and shadcn/ui component patterns
- TanStack Table for admin data views
- Recharts for dashboard analytics
- Server-Sent Events for live notifications

Figure 9 Notifications Page

3.1.5 Research Data Scope of UX

The UX scope focuses on reducing friction in reporting and searching items. Users complete guided forms with validation, upload up to five images, filter listings by type/category/status, and receive match suggestions based on category, building, and title similarity.

Core entities include User, Item, Claim, and Notification. Item records store type (LOST/FOUND), status (OPEN/RESOLVED/CLOSED), category, building, room hint, event date, images, and contact preferences. Claim records store claim type (FINDER/OWNER), proof message, proof images, and review status (PENDING/APPROVED/REJECTED).

Figure 10 Database Diagram

3.2 Data Collection Instrument

3.2.1 Data Collection Procedure

- Staff interviews on current lost-and-found handling and claim verification.
- Survey questionnaires for students and staff on reporting difficulty and recovery experience.
- Direct observation of manual reporting and notice-board practices.
- Document analysis of existing notice records and informal social media posts.

3.2.2 Statistical Data

Question 1: How old are you?

Figure 11 Respondents of question 1

66.7% of respondents were aged 18–25, indicating a primarily student population.

Question 2: What is your gender?

Figure 12 Respondents of question 2

58.3% male and 41.7% female participation.

Question 3: What is your role on campus?

Figure 13 Respondents of question 3

70% students, 20% staff, and 10% admin/support.

Question 4: Have you ever lost or found an item on campus?

Figure 14 Respondents of question 4

83.3% answered Yes, confirming high relevance.

Question 5: How do you currently report or search for lost items?

Figure 15 Respondents of question 5

75% use informal channels; only 25% use an official process.

Question 6: What challenges do you face?

Figure 16 Respondents of question 6

Manual search (75%), delayed response (58.3%), lack of photos (41.7%), no claim tracking (33.3%).

Question 7: How often do you fail to recover a lost item?

Figure 17 Respondents of question 7

41.7% sometimes, 25% frequently, showing a major service gap.

Question 8: How effective is the current process?

Figure 18 Respondents of question 8

Only 33.3% rated it very effective; most selected neutral or low ratings.

Question 9: Would image uploads help verification?

Figure 19 Respondents of question 9

91.7% answered Yes.

Question 10: Would searchable online listings help?

Figure 20 Respondents of question 10

88.3% answered Yes.

Question 11: Are staff willing to review digital claims?

Figure 21 Respondents of question 11

83.3% answered Yes.

Question 12: What is the expected recovery improvement?

Figure 22 Respondents of question 12

Most expect moderate to high improvement with BLFMS.

Question 13: Main advantages of BLFMS?

Figure 23 Respondents of question 13

Faster matching, better proof, centralized records, and notifications.

Question 14: Satisfaction with current system (1–5)?

Figure 24 Respondents of question 14

58.3% rated 1–3, indicating dissatisfaction.

Question 15: Would you recommend BLFMS?

Figure 25 Respondents of question 15

91.7% answered Yes.

3.3 Sampling Technique

Purposive sampling was used to include students, front-desk staff, and administrators directly involved in or affected by lost-and-found processes. A total of 30 respondents were selected.

3.4 Validity and Reliability

3.4.1 Validity

Content validity was ensured by aligning questionnaire items with research objectives and reviewing them for clarity and relevance.

3.4.2 Reliability

Cronbach's Alpha was used to test internal consistency. All major factors achieved values of 0.70 or above, indicating acceptable reliability.

Variable / Factor	Number of Items	Cronbach's Alpha	Interpretation
Reporting Experience	5	0.79	Reliable
Search and Recovery	5	0.76	Reliable
Claim and Verification	5	0.81	Reliable
System Performance	5	0.86	Highly Reliable
Overall Questionnaire	20	≥ 0.70	Acceptable Reliability

Table 3 Reliability Test Results (Cronbach's Alpha)

CHAPTER 4: DATA ANALYSIS

This chapter presents the analysis of data collected from questionnaires, interviews, and observations conducted at Beltei International University. The analysis focuses on identifying the strengths and weaknesses of the current lost-and-found practices and proposes suitable solutions.

4.1 Analysis of the Strengths

4.1.1 Effectiveness of the Current Lost and Found Practices

The survey results indicate that the current lost-and-found practices demonstrate a basic level of effectiveness. A proportion of respondents (33.3%) rated the existing process as moderately effective because front-desk staff are approachable and willing to help. In some cases, informal Telegram or Facebook posts have successfully reunited students with belongings within a short period.

Aspect Evaluated	Survey Result	Interpretation
Overall process effectiveness	33.3% rated effective	Basic support exists but is inconsistent
Staff willingness to help	High	Front-desk support is a key strength
Informal channel success	Occasional quick recovery	Works only for visible/high-traffic cases
Record keeping	Partial written logs	Some accountability but not searchable

Table 4 Effectiveness of Current Lost and Found Practices

4.1.2 Communication and Awareness Initiatives

Another strength is the active communication culture among students and staff. Findings show that 66.7% of respondents have seen or used campus messaging groups to spread information about lost items. Staff also display notices at the student center and front desk.

Indicator	Percentage	Interpretation
Use of messaging groups	66.7%	Fast but unstructured communication
Front-desk notice posting	50.0%	Visible to walk-in visitors only
Peer-to-peer sharing	75.0%	Community-driven but unreliable

Table 5 Communication and Awareness Initiatives

4.1.3 Monitoring and Reporting Practices

Some administrative monitoring exists through handwritten logs and end-of-week summaries. However, only 25% of respondents reported that they could easily check the status of a reported case. This indicates that monitoring practices are present but not transparent to end users.

4.2 Analysis of Weaknesses

4.2.1 Dependence on Manual and Fragmented Processes

The most significant weakness is reliance on manual and fragmented channels. 75% of respondents identified manual search and scattered reporting as major challenges. Users must check multiple places—front desk, social media, and class group chats—with no guarantee that information is current.

4.2.2 Poor Searchability and Incomplete Item Details

58.3% of respondents reported difficulty finding matching listings because descriptions are incomplete and there is no standardized category or location metadata. Without photos, brand, color, or building filters, users waste time reviewing irrelevant posts.

4.2.3 Delayed Communication and Weak Claim Verification

Delayed response was reported by 58.3% of respondents. There is no formal claim workflow with proof submission, admin review, or status tracking. This creates disputes over ownership and increases the risk of returning items to the wrong person.

4.3 Solution(s) Dealing with the Weaknesses

4.3.1 Implementation of BLFMS

To address the identified weaknesses, the implementation of BLFMS is strongly recommended. The system provides structured lost/found forms, searchable listings, image galleries, match suggestions, claim submission, and admin approval. Automated validation through Zod schemas ensures complete and accurate reports.

4.3.2 Integration, Training, and Real-Time Notifications

BLFMS integrates NextAuth authentication, Cloudinary image storage, Prisma data access, and REST APIs documented in OpenAPI. Real-time notifications through Server-

Sent Events inform users when claims are submitted or reviewed. Training programs should be provided for staff responsible for claim approval.

4.3.3 System Monitoring, Customization, and Feedback Mechanisms

Admin dashboards with charts for items over time, category distribution, and resolution rates support continuous monitoring. A feedback channel should allow users to suggest improvements. Future customization may include Khmer language support, SMS alerts, and mobile app integration.

Research Hypotheses:

H1: BLFMS significantly improves the accuracy of lost-and-found reporting.

H2: BLFMS significantly improves search and recovery efficiency.

H3: BLFMS significantly improves claim traceability and verification.

H4: BLFMS significantly improves communication timeliness.

H5: BLFMS has a positive impact on user satisfaction.

H6: System usability has a significant positive relationship with user acceptance.

CHAPTER 5: RESEARCH FINDINGS AND DISCUSSIONS

This chapter presents the key findings derived from the data analysis and discusses them in relation to the research objectives and conceptual framework.

5.1 Research Findings

5.1.1 Findings on Current Lost and Found Practices

The findings reveal that current practices rely heavily on manual and informal channels. 75% of respondents indicated that reporting and searching are time-consuming and inefficient. Additionally, 58.3% experienced delayed responses or no follow-up after reporting an item. Although staff are willing to help, the absence of a centralized digital system limits operational effectiveness.

5.1.2 Findings on System Efficiency and Accuracy

Accuracy and efficiency are major concerns. Only 16.7% of respondents believed the current process never fails to recover items, while the majority reported occasional or frequent failure. The lack of photo evidence, category filters, and match suggestions contributes significantly to these inefficiencies.

5.1.3 Findings on User Perception and Readiness for BLFMS

There is strong support for BLFMS implementation. 91.7% of respondents agreed that a digital platform would improve recovery speed and claim transparency. 88.3% said searchable online listings would help, and 91.7% would recommend implementing the system.

5.2 Discussion of Findings

5.2.1 Impact of Manual Processes on Campus Service Management

The findings confirm that manual lost-and-found processes negatively impact service efficiency and user satisfaction. Fragmented communication channels increase the time required to match owners with found items and create uncertainty around case status.

5.2.2 Role of BLFMS in Enhancing Efficiency and Accuracy

BLFMS acts as the intervening system that transforms reporting inputs into reliable service outputs. Features such as validated forms, image uploads, match confidence

scoring, claim review, and live notifications directly address the weaknesses identified in Chapter 4.

5.2.3 Implications for Campus Performance and Student Satisfaction

Improved lost-and-found management has a positive impact on student satisfaction and campus reputation. Faster recovery reduces stress and financial loss. Administrators gain measurable insight into open cases, resolution rates, and claim outcomes through dashboards and reports.

CHAPTER 6: CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

The research at Beltei International University demonstrates that lost-and-found management remains a high-impact campus service with significant room for digital improvement. Manual and informal methods create delays, search friction, and weak claim accountability. The proposed BLFMS offers a practical, technically sound, and user-supported solution. With structured reporting, image evidence, match suggestions, notifications, and admin review tools, the system can improve recovery efficiency and campus service confidence.

6.2 Recommendations

- Deploy BLFMS as the official campus lost-and-found platform.
- Train front-desk staff and administrators on claim review and item lifecycle management.
- Promote the platform during orientation and through faculty announcements.
- Maintain data privacy controls for contact details and proof images.
- Monitor KPIs such as open cases, resolution time, and claim approval rate.
- Extend the system with email/SMS alerts and a dedicated mobile application.
- Conduct annual usability reviews and feature updates based on user feedback.

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APPENDICES A: Geography

[Insert map of Cambodia and BIU campus location]

APPENDICES B: Data Structure and Frameworks

Database entities: User, Item, Claim, Notification, Account, Session. Key enums:

ItemType (LOST, FOUND), ItemStatus (OPEN, RESOLVED, CLOSED), ClaimType (FINDER, OWNER), ClaimStatus (PENDING, APPROVED, REJECTED).

APPENDICES C: System Interface (UI)

[Insert screenshots: Home, Dashboard, Report Lost, Report Found, Item Detail, Claims, Admin Dashboard, API Docs]

APPENDICES D: Survey Questionnaires in Google Form

[Insert full Google Form questionnaire for lost-and-found survey]

APPENDICES E: Activities And Planning

Week 1: Requirement gathering and problem identification

Week 2: Literature review and conceptual framework

Week 3: Database schema and UI wireframes

Week 4: Authentication and item reporting modules

Week 5: Claim workflow and notifications

Week 6: Admin dashboard, API docs, testing, and thesis writing

APPENDICES F: Photos

[Insert development progress photos and deployment screenshots]

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